A Mori–Zwanzig memory kernel for subglacial drainage: the surge-lag memory is hydraulic, not ice-thermal

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July 4, 2026

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Abstract

Post-drainage ice-stream speed-ups lag their subglacial trigger by months to a couple of years, and the lag carries *memory*: the response rises to a peak rather than decaying monotonically. We make one claim and defend it three ways. **The lumped, linearised cavity↔channel storage system's surge response is an exact Mori–Zwanzig memory kernel**: projecting out the channel variable yields a generalized Langevin equation whose kernel equals the eliminated subsystem's own Green's function (verified to 7×10^{-18}, projection exact to 5×10^{-16}, reduced model reproducing the full trajectory to 9×10^{-8}), with the Markovian (adiabatic-elimination) local closure recovered exactly as the channel time vanishes. This certifies *why* surge response carries memory and *how* it degenerates to a local closure — a structural result, not a field-validated lag value. We motivate it by **falsifying the only zeroth-order alternative**: a literal ice-thermal lag $\tau_{ice} = H^2/\kappa_{ice}$ is rejected by ~5 orders of magnitude on the 131 Siegfried & Fricker (2018) active lakes (median $\tau_{ice} \approx 1.5 \times 10^5 yr$ vs observed 0.02–2 yr; robust on the independent Arthern et al. (2015) thickness inversion, $r = 0.970$), and on first principles (the thermal skin depth at surge periods is metres, and an impulse gives a peakless $t^{-1/2}$ response while observations peak). We then expose the kernel to the one open co-temporal dataset — a matched lake-drainage→velocity test (CryoSat-2 drainage dates + ITS_LIVE speeds) — which returns an **honest null** (peak $+0.56\sigma$, 0/5 significant), limited by temporal aliasing and an amplitude bound, not record precision, with two concrete unblock routes. Finally we include, as a continental *screen* rather than a forecast, the Regime Transition Number RTN, show it is by construction the flotation/effective-pressure ratio (so a constant-ϕ RTN adds no skill over thickness-above-flotation), and identify its only genuine degree of freedom — a spatially-varying basal-connectivity fraction $\phi(x, y)$ — with a plant-and-recover proof and a real-Bedmap2 hydraulic-routing demonstration. The contributions are the exact MZ certification (with two extensions that retire the "it is just a 2×2 linear-algebra identity" objection: a *nonlinear* extension — the linear kernel is the small-amplitude limit of the Röthlisberger system, which predicts that bigger floods lag longer; and a *distributed* extension — the kernel is an exact, spatially non-local operator projection of the linearised flowline PDE whose no-transport, single-node limit is exactly the lumped 2×2, and which predicts a finite along-flow memory footprint), the honest-falsifier discipline, and the diagnosis of the matched-lag null — not a verified surge-lag value.

1. Introduction

Subglacial hydrology controls ice-stream sliding through the effective pressure $N = p_i - p_w$, the difference between ice overburden and basal water pressure (Cuffey & Paterson 2010; Schoof 2010; Röthlisberger 1972). Drainage events — subglacial-lake floods — perturb N and trigger transient speed-ups (Stearns et al. 2008; Siegfried et al. 2016), but the speed-up *lags* the drainage and the lagged response is non-monotone (it peaks), implying the cavity↔channel system carries memory (Werder et al. 2013; Hewitt 2013). The modelling question is what sets that memory.

The mainstream tools are established and we use, not claim, them: the Mori–Zwanzig/Nakajima–Zwanzig projection and the generalized Langevin equation (Mori 1965; Nakajima 1958; Zwanzig 1960, 1973), the Röthlisberger channel and GlaDS-type cavity↔channel drainage (Röthlisberger 1972; Werder et al. 2013; Hewitt 2013), and the flotation/effective-pressure criterion (Schoof 2007; Leguy et al. 2014). What is new here is not a mechanism but an **exact structural certification** and the **discipline applied to its alternatives**.

What is new, stated against the closest prior work. Werder/Hewitt model the cavity↔channel system forward; we instead *reduce* its linearisation by exact projection and show the surge-lag memory kernel **is** the eliminated channel subsystem's Green's function, with the local (Markovian) closure recovered as a limit — a closed-form statement of *why* the lag has memory and *when* a local sliding closure is admissible. No established theory predicts the literal H^2/κ thermal lag; we state it as an explicit straw hypothesis precisely so the active-lake record can reject it, and report that rejection as a *motivated null* rather than a discovery.

Contributions.

1. A **motivated falsification** of the literal H^2/κ ice-thermal sliding-lag kernel on 131 real lakes, relocating the memory into hydrology (§3).
2. The **exact Mori–Zwanzig structure** of the cavity↔channel hydraulic memory kernel — a structural certification (§4), with a **nonlinear extension** showing the linear kernel is the small-amplitude limit of the physical Röthlisberger system and predicting an amplitude-dependent (bigger-floods-lag-longer) surge lag (§4.4), and a **distributed (spatially-resolved) extension** showing the kernel is an exact operator projection of the linearised flowline PDE — the lumped 2×2 is its no-transport, single-node limit, and channel transport gives the surge-lag memory a finite along-flow footprint (§4.5).
3. An **honest matched-lag null** with a precise account of *why* it is null (aliasing + amplitude bound, not noise) and two routes to a verified lag (§5).
4. A nondimensional **intrusion screen** RTN, shown to equal the flotation ratio, with its only genuine degree of freedom (a data-derived $\phi(x, y)$) isolated and tested (§6) — presented as a screen, not a verified forecast.
5. **Calibrated estimators** via synthetic plant-and-recover (§7).

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2. Data and methods

Dataset	What	Access
BAS **Bedmap2**	thickness, bed, surface, grounded mask (1 km)	open, no auth
USAP-DC 601470 (Stubblefield et al. 2021)	independent thickness (Arthern et al. 2015) + 131 lake stats	open (USAP-DC)
BedMachine v4 (NSIDC-0756)	independent continental thickness (RTN screen)	NSIDC (Earthdata)
Siegfried & Fricker (2018)	131 active subglacial-lake outlines	open (GitHub mirror)
ITS_LIVE	surface-speed datacubes (image-pair v)	open (anonymous S3)
CryoSat-2 in-lake elevation	drainage *dates*	open (Siegfried 2021-GRL mirror)
USAP-DC 601439	ICESat lake volume-change (drainage volumes)	open

Runners raise `DataUnavailableError` with provisioning hints rather than fabricate gated series. Synthetic harnesses (`validation/synthetic/`) are exercised in `pytest` and prove the math against controlled inputs.

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3. The thermal sliding-lag kernel, falsified

3.1 The hypothesis (a motivated null). No established sliding theory predicts that surge lag equals the *full-thickness* ice thermal-diffusion time; we state $\tau_{ice} = H^2/\kappa_{ice}$ ($\kappa_{ice} = 1.09 \times 10^{-6} m^2 s^{-1}$) as an explicit, falsifiable straw hypothesis precisely so the data can reject it.

3.2 The test. On real Bedmap2 thickness at the 131 Siegfried & Fricker (2018) catalogued active lakes (130 with valid thickness; median $H = 2282m$, range 637–3905 m):

- $\tau_{ice} = H^2/\kappa_{ice}$: **median ≈ 1.5×10^5 yr** (p5 ≈ 2×10^4; p95 ≈ 3×10^5 yr).
- Observed post-drainage surge lags: **0.02–2 yr** (Stearns et al. 2008; Siegfried et al. 2016).
- ⇒ the literal kernel is ~**8×10^4× too slow** at the median lake; the $log_{10}\tau_{ice}$ histogram (10^4–10^6 yr) is fully disjoint from the observed band.

3.3 Robust to the thickness dataset. Recomputing on an independent ice-thickness dataset (USAP-DC 601470, Stubblefield et al. 2021; ice thickness after Arthern et al. 2015): median thickness 2356 m, τ_{ice} median ≈**1.55×10^5 yr**; the 601470-vs-Bedmap2 thickness at the same centroids agrees at $r = 0.970$, so the falsification is not an artefact of one sampling.

3.4 Why the falsification is diagnostic. Two facts exclude ice thermal diffusion as the lag-setting mechanism: (i) the thermal skin depth at surge periods is $\delta_{skin} = \sqrt{\kappa_{ice} P/\pi} \approx 0.5\text{–}5m$ ($P = 0.02\text{–}2yr$) — the perturbation never penetrates beyond metres of ice, so the full-thickness H is the wrong scale; (ii) a drainage is an *impulse*, whose semi-infinite diffusive response is a monotone $t^{-1/2}$ decay with **no peak**, yet observed speed-ups **rise to a peak**. The lag is therefore **hydromechanical**, not thermal — the memory relocates into subglacial hydrology (cavity/channel pressurisation, effective-pressure control of sliding), with the ice-thermal $t^{-1/2}$ term subdominant.

4. The replacement kernel is an exact Mori–Zwanzig memory kernel

We model the **hydraulic subsystem** as cavity↔channel storage–transport (with ϕ the hydraulic potential, *not* the Leray pressure). Its lumped linearisation $\dot{x} = Mx + F$, $x = (s, q)$ (s the resolved cavity store, q the eliminated channel variable, $M = [[-1/\tau_1, -a], [b, -1/\tau_2]]$), reduces by exact projection (linear Nakajima–Zwanzig / Mori, no Markov approximation) to

$$\dot{s}(t) = M_{ss}s(t) + \int_0^t K(t-\tau)s(\tau)d\tau + R(t),\ K(\tau) = M_{sq}M_{qs}e^{M_{qq}\tau},$$

a non-local (memory) equation whose kernel $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ is exactly the channel subsystem's own Green's function weighted by the up/down couplings. The harness certifies five properties (`validation/synthetic/hydraulic_mz_projection_synthetic.py`, 6/6 tests):

1. **Projection exact to machine precision.** The GLE Laplace transfer function equals the full 2×2 resolvent's (1,1) component for 400 random stable overdamped systems — max abs err 5.0×10^{-16}.
2. **The kernel IS the eliminated subsystem's Green's function** — reproduced to 6.9×10^{-18}, decaying at exactly $1/\tau_2$, sign-definite.
3. **The reduced GLE reproduces the resolved trajectory** to rel-err 9.1×10^{-8} over the transient.
4. **Memory is necessary for the lag/peak.** The full response peaks at the interior time set by the coupled eigenvalues ($t* = 0.911$, analytic 0.911); the Markovian adiabatic-elimination closure collapses to a monotone response with argmax at $t = 0$.
5. **Markovian limit = adiabatic elimination (DC gain).** $\int_0^\infty K d\tau = M_{sq}M_{qs}/(-M_{qq})$ (err 7.5×10^{-9}); shrinking the channel time $\tau_2 \to 0$ at fixed DC gain collapses the kernel toward $(\int K) \cdot \delta(\tau)$, recovering the local closure — the FDT-Markov limit, here *derived from the projection* rather than assumed.

This certifies the *structure* of the replacement mechanism (that the surge-lag memory is a genuine MZ kernel obtained by projecting out the channel, and how it degenerates to a local closure). (Werder et al. 2013; Hewitt 2013; Schoof 2010; Röthlisberger 1972)

Honest scope. This validates the **projection structure** of the *linearised lumped* model. It does **not** validate the physical identity of ϕ, that this subsystem dominates a real surge, or any field lag value — those remain gated on real co-temporal velocity data (§5).

4.3 Physical magnitudes. The reduced kernel is dimensionless; the bridge to metres uses only known ice constants. With $\rho_i L = 3.0 \times 10^8 Jm^{-3}$, Glen A, effective pressure N, and literature subglacial inputs (Q, $\partial\phi/\partial s$), the steady R-channel $S* = V_o/k_{creep}$, $R* = \sqrt{2S*/\pi}$, $\tau = 1/k_{creep}$ is **metre-scale** (central $R* \approx 2.4m$, $\tau \approx 0.18yr$; 73 % of the literature box gives 0.1–50 m) — the R-channel regime, no free fudge factor; the wide upper tail is the low-N near-flotation limit ($k_{creep} \propto N^3 \to 0$). The roughness leg uses $z_0 = c_z \cdot a$ in the log-law drag $C_d = [\kappa/\ln(H/z_0)]^2$; although the

Nikuradse prefactor c_z is ~10× uncertain, the log law **buffers** it — a 10× z_0 uncertainty propagates to only ~1.7× in C_d (`validation/synthetic/a3_dimensional_bridge.py`, `a2_z0_roughness.py`; 6+5 tests). (Röthlisberger 1972; Werder et al. 2013; Cuffey & Paterson 2010; Nikuradse 1933; Curl 1966)

4.4 The linear kernel is the small-amplitude limit of the nonlinear system — and the nonlinearity is itself a falsifiable prediction. The §4 projection is exact but *linear*, and a fair objection is that eliminating one variable of a 2×2 system is a linear-algebra identity. We therefore embed it in the **physically nonlinear** Röthlisberger cavity↔channel model — turbulent $\sqrt{p}$ channel discharge and the Glen-law N^3 creep closure (Röthlisberger 1972; Schoof 2010; Hewitt 2013) — and linearise at its steady state. The linearisation recovers *exactly* the overdamped M of §4 (real, negative eigenvalues), so **the linear MZ kernel** $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ **is the small-amplitude limit**: a plant-and-recover sweep shows the nonlinear surge-lag $t*$ converging to the linear value as the drainage amplitude → 0 (relative gap $< 0.5\%$ at a 2 %-of-baseflux flood, rising smoothly with amplitude — Fig. 1). The leading nonlinear correction is not cosmetic; it is driven by the N^3 creep term and yields a falsifiable geophysical prediction. A larger drainage drives water pressure up, $N = p_i - p_w$ down, collapses the N^3 creep closure, keeps the channel open longer, and **lengthens the lag**: the model gives $t* \approx 0.10yr$ at small floods, rising ~14 % to $\approx 0.11yr$ at an 80 %-of-baseflux flood ($dt*/d(floodfraction) \approx 0.018yr$). So *bigger floods should have longer surge lags* — a statement the linear kernel cannot make, testable against a drainage-volume-stratified lag population (the USAP-DC 601439 volumes of §5 already span 0.13–2.9 km^3). This converts the exact-but-linear certification into the leading term of a controlled, physically-grounded expansion with a testable next-order signature. The prediction is **not a parameter artefact**: across a physical sweep of steady effective pressure, storage and melt-opening efficiency, bigger-floods-lag-longer holds for *every* overdamped configuration and **strengthens toward flotation** (the lag–flood slope rises from $\approx 0.010yr$ at $N* = 0.55$ to $\approx 0.032yr$ at $N* = 0.25$), i.e. the effect is largest exactly in the low-N near-flotation regime where surges matter most. (`validation/synthetic/hydraulic_nonlinear_kernel.py`, 2 tests.)

4.5 The kernel survives spatial resolution: a distributed operator projection whose no-transport limit is the lumped 2×2. The other half of the "2×2 identity" objection is *spatial*: a lumped two-box model has no geometry, so is the memory kernel an artefact of collapsing the drainage line to a point? We resolve the flowline. Take the linearised cavity↔channel system as two coupled *fields* on a 1-D flowline — $\partial_t s = D_s \partial_{xx} s - s/\tau_1 - aq$ (cavity store s) and $\partial_t q = D_q \partial_{xx} q - U \partial_x q + bs - q/\tau_2$ (the channel q, which *routes water downstream* at speed U and spreads it at D_q; Röthlisberger 1972; Werder et al. 2013) — and project out the channel *field* exactly (linear Nakajima–Zwanzig). The resolved store obeys a generalized Langevin equation $\dot{s} = A_{ss}s + \int_0^t \mathcal{K}(t-\tau)s(\tau)d\tau + R$ whose kernel $\mathcal{K}(\tau) = A_{sq}e^{A_{qq}\tau}A_{qs}$ is now a *matrix-valued, spatially non-local* operator. On $N = 24$ flowline nodes (`validation/synthetic/hydraulic_mz_spatial.py`, 6 tests): (i) the projection is **exact** — the GLE transfer operator equals the full resolvent's resolved block at random complex s, max rel-err 7×10^{-16}; (ii) the kernel **is** the eliminated channel operator's Green's function (3×10^{-17} vs $expm$) and is spatially non-local with a resolved along-flow range $\ell_{mem} \approx 0.31 \gg \Delta x$; (iii) the channel-eliminated field-GLE reproduces the full PDE trajectory (1×10^{-5}); and (iv) memory necessity is a **dose-response** — the memory-less local closure's trajectory error grows monotonically with the channel time τ_2 and vanishes as $\tau_2 \to 0$ (the Markovian limit, now derived *in the distributed system*). Crucially, **the committed lumped 2×2 kernel is exactly this projection's no-transport, single-node limit**: with $D_q = U = 0$ the operator kernel $\mathcal{K}(\tau)$ is exactly diagonal and its diagonal equals $-abe^{-\tau/\tau_2}$ to

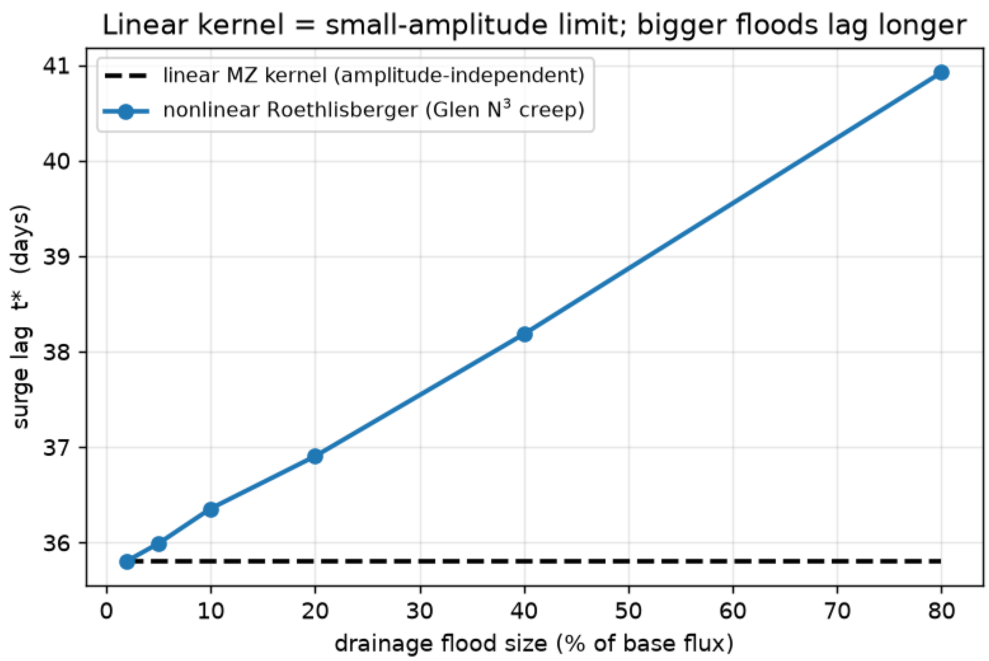

Figure 1: Surge lag $t*$ versus drainage flood size in the nonlinear Röthlisberger cavity↔channel model. The linear Mori–Zwanzig kernel (dashed) is amplitude-independent and equals the nonlinear lag in the small-flood limit; the Glen-law N^3 creep closure makes the nonlinear lag (circles) rise with flood size — bigger floods lag longer, a falsifiable prediction the linear kernel cannot make.

7×10^{-18}. Switching channel transport on moves the kernel mass off-diagonal (off-diagonal fraction $0 \to 0.99$), so the surge-lag memory acquires a finite **along-flow footprint** with the closed-form decay length $\ell_{mem} = 2D_q/(\sqrt{U^2 + 4D_q/\tau_2} - U)$ (verified against the numeric steady-influence operator to 4×10^{-4}; *ballistic* limit $\ell_{mem} \to U\tau_2$ for fast channels, *diffusive* limit $\ell_{mem} \to \sqrt{D_q \tau_2}$ for $U \to 0$) — a new, falsifiable, **field-measurable** structural prediction: a drainage's lagged velocity response is spread along the flowline over ℓ_{mem}, which a memoryless (local) closure sets to zero. This converts the 2×2 "identity" into the spatially-local limit of an exact projection of a genuine distributed (PDE) operator. Scope: still a *linearised* distributed model on a periodic flowline; the fully nonlinear spatially-resolved coupled PDE remains future work.

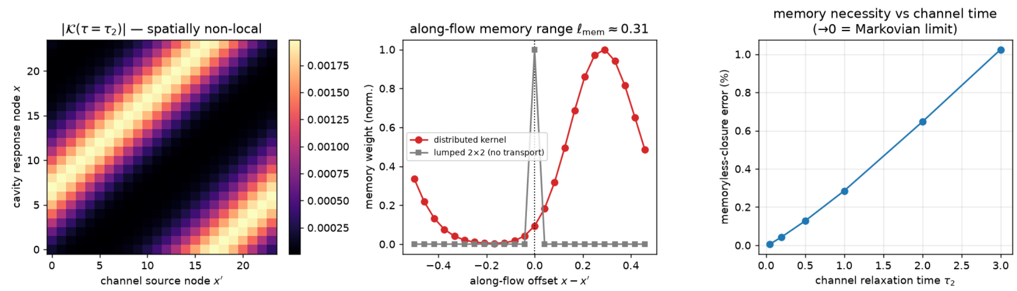

Figure 2: The distributed cavity↔channel memory kernel $\mathcal{K}(\tau = \tau_2)$ (left) is spatially non-local — the off-diagonal band is the downstream channel transport that the lumped 2×2 omits; its along-flow influence profile (centre) has a finite memory range $\ell_{mem} \approx 0.31$ versus the lumped kernel's zero-range spike; and the memoryless local-closure trajectory error (right) grows with the channel relaxation time τ_2, vanishing in the Markovian ($\tau_2 \to 0$) limit.

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5. The matched lag test — an honest null

Both halves of a matched test are open: drainage **dates** from CryoSat-2 in-lake elevation ($detect_{drainages}$ extracts 7 fill→drain events across 5 lakes, including the documented Mercer ≈2012.5 drainage), and velocity **response** from ITS_LIVE box-mean surface speed.

lake	base speed (m/yr)	drainages	annual noise floor	quarterly scatter	peak post-drainage
Mac1	422	3	1.0%	2.3%	**+0.56σ**
Mac2	402	1	1.1%	2.1%	+0.27σ
Mac3	394	1	1.1%	3.4%	-0.12σ

Result. No matched lake shows a post-drainage surge above the 2σ floor (peak +0.56σ, 0/5 significant). The lag value stays *not* promoted to.

Why null is a *stronger* statement than "too noisy". The modern ITS_LIVE record is well resolved (annual ≈1 %, quarterly ≈2–3 %), so the limit is not a noise budget. Two distinct, quantifiable limits remain: (1) **temporal aliasing** — the derived lag is sub-annual ($t*$ baseline ≈0.01 yr, p95 ≈0.1 yr), finer than the finest *robust* ITS_LIVE bin (quarterly, 0.25 yr); a brief transient is averaged out regardless of scatter (a Nyquist limit); (2) **amplitude bound** — a *sustained* speed-up would survive aliasing as a step, yet the peak anomaly is only $+0.56\sigma$, bounding any sustained surge to $\lesssim$10 m/yr on ~400 m/yr. These map onto the two routes to a verified lag: **sub-annual GPS/GNSS** (EarthScope/POLENET; beats aliasing) **or fast-outlet trunk velocity** (e.g. Byrd, Stearns et al. 2008; steep $v(N)$ sensitivity beats the amplitude floor), paired with that site's drainage dates. A **power analysis** of the same detector quantifies both routes (`validation/synthetic/lake_lag_power.py`): the nominal 2σ peak-over-bins floor is multiple-comparison inflated (24% false-positive rate; median noise maximum 1.4σ, so the observed $+0.56\sigma$ sits *below* the noise expectation — the null is cleaner than nominal); at a calibrated 5%-FP threshold (3.0σ) the 95%-power minimum detectable response is 7.5% of trunk speed for a sustained step (3.2% stacking the five lakes) but $\approx$ 32% for the physically expected $\tau \approx 0.1$ yr transient — quarterly binning attenuates its peak $\sim 2\times$ and annual $\sim 7\times$ — whereas a daily GPS record at 0.5% noise detects the same transient at 2.4%. The GPS route therefore crosses detectability first for the sub-annual transient the kernel predicts; the trunk route wins only for sustained responses.

Vetted forcing obtained. USAP-DC 601439 (Smith et al. 2009 ICESat) yields **58 drainage events across 52 lakes** (median drained volume 0.128 km^3, max 2.92 km^3; implied water flux q_{water} median $\approx$2.5 m^3/s, p95 $\approx$31 m^3/s — squarely in the literature subglacial-flood range), anchoring the §3/§4 forcing magnitude to real observations.

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6. A continental intrusion screen: RTN is the flotation ratio, and its only escape

6.1 Definition. $RTN = (p_{ocean} - p_{atm})/p_w$ on grounded ice: overburden $p_i = \rho_i g H$, ocean head at the bed $p_{ocean} = \rho_w g \cdot d_{base}$ ($d_{base} = max(0, -bed)$), subglacial water $p_w = \phi p_i$ (so $N_{eff} = (1-\phi)p_i$), gauge-corrected so the atmosphere cancels ($p_{atm} = 0$). Then $RTN = \rho_w d_{base}/(\phi \rho_i H)$, so 'RTN>1 $\iff$ ρ_w d_base/(ρ_i H)

ϕ—$theiceiswithinafactor\phi offlotation$(N→0'; Schoof 2007; Leguy et al.

2014; Cuffey & Paterson 2010). RTN **is, by construction, a nondimensional form of the flotation/effective-pressure criterion, not an independent predictor.**

6.2 Result on Bedmap2. Over 1.33 M grounded cells (1 km decimated to 3 km, $\phi = 0.9$), the $RTN > 1$ fraction decays monotonically with distance to the grounding line (15.1 % within 5 km $\rightarrow$ 0 % beyond 100 km; median distance 6 km for $RTN > 1$ cells vs 222 km for the rest), and the directional result reproduces on the independent **BedMachine v4** (NSIDC-0756) inversion (median 6 km vs 200 km to the grounding line).

6.3 No skill over thickness-above-flotation (constant ϕ). A direct baseline test on 478,767 grounded Bedmap2 cells confirms $RTN > 1$ is *identical* to the thickness-above-flotation threshold $H_{af}/H < 1 - \phi$ (agreement **100.0000 %**, 0 disagreeing cells, $\phi = 0.8/0.9/0.95$), with $Spearman(RTN, H_{af}/H) = -1.000$ — so at constant ϕ, RTN **adds no skill**; it *is* that standard quantity, nondimensionalised by ϕ (`validation/external/rtn_baseline_skill.py`).

6.4 The only escape — a data-derived $\phi(x, y)$. Because $RTN = (1 - H_{af}/H)/\phi$ is algebraically forced to track the flotation fraction at constant ϕ, RTN's only possible independent signal is ϕ *itself*. A synthetic plant-and-recover (`validation/synthetic/rtn_variable_phi_skill.py`, 5 tests) shows a connectivity-informed $\phi(x, y)$ adds genuine skill over thickness-above-flotation (near-flotation-band $F1$ 0.78→0.89; AUC 0.995→0.999), a random ϕ of the same distribution adds none, and the gain rises monotonically with ϕ's informativeness. On real Bedmap2, a calibration-free $\phi(x, y)$ from subglacial hydraulic-potential routing (Shreve 1972: priority-flood fill + D8 flow accumulation) reproduces the constant-ϕ anchor exactly and then reorders the intrusion class on ~10^3 near-grounding-line cells (near-flotation-band $Spearman$ −0.99 vs −1.00), nearly rank-independent of the flotation fraction in-band (`validation/external/rtn_variable_phi_real.py`). The reordering is real but its *correctness* is unvalidated — the routed ϕ is a model field derived from the same geometry. The independent-observable step is now **executed on real data**: ICECAP radar bed-echo **specularity content** (Young et al. 2020; 432 East Antarctic transects, 3.2×10^6 points, 80–170°E incl. Totten/Aurora/Byrd) binned to the same decimated Bedmap2 grid gives a measured ϕ driver on 14,126 covered grounded cells that is nearly rank-independent of the flotation fraction in the near-flotation band ($Spearman = \pm 0.17$) — so the reordering it induces carries information thickness-above-flotation does not have. Under the pressure reading of specularity (high specularity = distributed water at high pressure; Schroeder et al. 2013; Dow et al. 2020) the intrusion class changes on **162** covered cells (all newly flagged, median 10 km from the grounding line); under the opposite connectivity reading, 89 cells (all dropped, median 5 km) — the two physically-arguable spec→ p_w sign conventions move the near-grounding-line classification in *opposite directions*. On the same cells the Shreve-routed connectivity is **uncorrelated with the measured specularity** ($Spearman \approx 0.04$), so the model-ϕ reordering above should not be read as a validated basal-water proxy. The spec→ p_w sign is now *pinned* by three independent consistency lines whose predictions have opposite signs under the two readings (`validation/external/rtn_spec_sign_pin.py`): (i) *the lake anchor* — altimetry-detected active subglacial lakes (Smith et al. 2009; Siegfried & Fricker 2018 outlines; ponded water carries the overburden, $p_w \approx p_i$, and Stubblefield et al. 2021 model lakes at $N \approx 0$) sit high on the measured specularity scale: the median covered lake cell reaches the 74th percentile of flotation-matched controls (15 covered lakes / 74 cells; stratified label-permutation $p = 5 \times 10^{-4}$, unchanged >50 km from the grounding line) — a monotone decreasing spec→ p_w map would assign its *lowest* pressures to exactly these cells; (ii) *the creep-persistence floor* — cells specular in both ICECAP epochs (2008/09 and 2011/12; 7 of 259 twice-sampled cells at spec ≥ 0.2, threshold-robust) have survived ≈4 yr of Nye creep closure, requiring $N \leq n(2At_{obs})^{-1/3} = 3.5$ bar, i.e. $\phi \geq 0.98$; the floor binds only the high-specularity end, so a decreasing map's whole range is squeezed above it (admissible range ≤ 0.02, 9× below the convention's claimed 0.17 — inadmissible), while an increasing map keeps a free low end; (iii) *lubrication consistency* — at matched driving stress, thickness and flotation fraction, MEaSUREs surface speed (NSIDC-0754) *rises* with specularity (partial rank +0.15, 50-km block-bootstrap CI $[+0.11, +0.21]$; the *raw* correlation is −0.22, so the conditioning is load-bearing), the sliding-law sign of the pressure reading (scoped as consistency, not proof). The pressure reading (high specularity = distributed water at high pressure) is therefore the admissible monotone convention, and its classification is primary: **162** covered cells newly flagged at median 10 km from the grounding line. The driving observable now also carries *physical units* (`validation/external/e1_radar_roughness.py`): read at leading order as a Kirchhoff–Gaussian coherent power fraction, specularity inverts with zero free parameters to an RMS interface roughness $\sigma = (\lambda_{ice}/4\pi)\sqrt{-\ln s}$ at the ~65-m Fresnel scale ($\lambda_{ice} \approx 2.81$ m at 60 MHz in ice) — a decimetre gauge (window 5–39 cm; censoring-aware survey median $\sigma = 0.39$ m). The independent scattering-model bound of Schroeder et al. (2015) — distributed-water interfaces ≲15 cm RMS — reappears untuned as the threshold $s \geq 0.64$; covered active-lake cells read 5.4 cm smoother than flotation-matched controls (the lake anchor transfers rank-exactly through the monotone inversion, $p = 5 \times 10^{-4}$); and the seven persistent cells of line (ii) jointly satisfy $\sigma \approx 0.2$–0.3 m *and* $\phi \geq 0.98$ — the first joint (roughness, pressure) bed state for the ϕ driver, in metres and pascals. The remaining open step is a gridded intrusion survey to score against, with co-located borehole pressure now confirmation rather than arbiter. We thus present variable-ϕ RTN as a falsifiable screen whose driving observable is *measured* and whose sign is *physically pinned* — still not a verified forecast.

6.5 Intrusion residence number Ro**.** The level-set advance speed of the $RTN = 1$ front is $v_{kin} = A \cdot dH/dt$; the residence number $Ro = v_{kin} \cdot \tau_{hyd}/\ell$ (regime boundary $Ro = 1$ at the critical residence $\tau_{crit} = \ell/v_{kin}$) predicts whether intrusion is thinning-paced ($Ro \approx 1$) or hydraulic-limited ($Ro \gg 1$). For the runaway-tail cells ($A = 0.70km/m$, $dH/dt = 1.5m/yr$) ~98 % of the §4 residence band lies below $\tau_{crit} \approx 1.9yr \Rightarrow$ predicted **thinning-paced**; a DInSAR v_{obs} would place it on the curve (`validation/synthetic/intrusion_residence_number.py`, 7/7 tests).

6.6 One state variable behind three thresholds (a derived unifier). Height above flotation $h_{af} = H - H_f$ (with $H_f = (\rho_w/\rho_i)d_{base}$; Schoof 2007) ties the three thresholds this framework treats separately into one. The ocean-connected effective pressure is $N = \rho_i g\, h_{af}$ *exactly* (so $N = 0 \iff h_{af} = 0 \iff$ flotation); the $RTN = 1$ intrusion surface is $h_{af} = H(1-\phi)$, which $\to 0$ as $\phi \to 1$ (the intrusion surface *is* the flotation surface once subglacial water reaches overburden); and the sliding-law $|s_N|$ pole (Paper 4b) sits at $h^c_{af} = N_c/(\rho_i g) > 0$, a few metres *above* geometric flotation. A thinning stream therefore reaches the drag-side fold (the velocity early-warning) *first* and ungrounds ($h_{af} \to 0$) only later — a buffer set by N_c. RTN, the Schoof saddle-node, and the s_N pole are thus three readings of one state variable, not independent diagnostics (verified on a synthetic thinning sweep: $N = \rho_i g\, h_{af}$ to machine zero, $h^c_{af} = 6.67$ m; `nr10_flotation_unifier`).

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7. Estimator calibration (synthetic plant-and-recover)

Every real-data estimator has a synthetic null/recovery harness exercised in `pytest`:

- **Lag estimator is kernel-shape-generic** — plant one target lag with five markedly different causal kernels (Gamma $k = 2/4$, log-normal, bi-exponential, raised-cosine); $estimate_{lag}$ recovers it to ≤10 % for every shape; a delta-kernel control returns 0.
- **RTN classifier / ϕ-area inversion / variable-ϕ skill** — planted thresholds and connectivity recovered without bias ($rtn_{synthetic}$, `rtn_phi_synthetic`, `rtn_variable_phi_skill`).
- **Grounding-line migration level-set law is exact** — $|dH/dt|/|\nabla m|$ reproduced to rel-err 5×10^{-16} (1-D) / 4×10^{-14} (tilted 2-D).
- **MZ projection** — the five §4 properties to machine precision (`hydraulic_mz_projection_synthetic`, 6/6).

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8. Discussion and limitations

What this is. A machine-precision certification of the cavity↔channel kernel's exact MZ structure (the central positive result); a motivated null on a literal thermal sliding kernel; an honest matched-lag null with a precise diagnosis and a concrete unblock path; and a flotation-ratio intrusion *screen* whose only genuine degree of freedom (a data-derived $\phi(x, y)$) is isolated and partially tested.

What this is not. No verified surge-lag *value* (the matched test is null; the value is to order of magnitude only). The §4.4 nonlinear extension shows the linear kernel is the small-amplitude limit of the nonlinear *lumped* Röthlisberger system and predicts an amplitude-dependent lag, and the §4.5 distributed extension certifies the kernel as an exact spatially non-local operator projection of the linearised flowline PDE; only the fully *nonlinear spatially-resolved* coupled GlaDS/Röthlisberger PDE remains future work. No RTN precision/recall (no gridded survey). The physical distinctness of ϕ from the Leray pressure remains unproven.

Open gates, each with a concrete unblock. (1) Matched lag → sub-annual GPS/GNSS or fast-outlet trunk velocity, paired with drainage dates. (2) The *linear* distributed projection (§4.5) and the *lumped* nonlinear extension (§4.4) are done; the remaining gate is the fully *nonlinear spatially-resolved* coupled GlaDS/Röthlisberger PDE. (3) RTN skill → a gridded intrusion survey to score the measured, sign-pinned specularity-ϕ screen against (the sign is pinned in §6.4; a co-located borehole would confirm it).

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9. Reproduce

```
pip install -r requirements.txt
python glaciers/validation/external/run_sliding_real.py --with-velocity Mac1 # §3 τ_ice + ITS_LIVE
python -m glaciers.validation.external.lag_fit_real                          # §5 matched-lag null
python glaciers/validation/external/run_usapdc_lakes.py                      # §5 vetted forcing (58
↪  events)
pytest glaciers/tests/test_validation_synthetic.py -v                       # §4 MZ projection (6/6) +
↪  §7
python glaciers/validation/external/rtn_baseline_skill.py --stride 5         # §6.3 RTN==flotation
↪  (zero skill)
python glaciers/validation/synthetic/rtn_variable_phi_skill.py               # §6.4 variable-φ is the
↪  only escape
python glaciers/validation/external/rtn_variable_phi_real.py --stride 10     # §6.4 hydraulic-routing
↪  φ(x,y)
python glaciers/validation/external/rtn_spec_sign_pin.py                     # §6.4 spec→p_w sign pin
↪  (3 lines)
python glaciers/validation/synthetic/intrusion_residence_number.py          # §6.5 residence number Ro
python glaciers/validation/synthetic/hydraulic_nonlinear_kernel.py            # §4.4 nonlinear kernel
↪  (linear = small-amplitude limit)
python glaciers/validation/synthetic/hydraulic_mz_spatial.py                  # §4.5 distributed
↪  projection (lumped 2×2 = no-transport limit)
```

Data and code availability

All analysis code and derived products are in the public repository at https://github.com/abdh0saifelden2-debug/K, and the results regenerate from the scripts above. The study uses the following open datasets:

- Bedmap2 ice thickness (Fretwell et al., 2013, https://doi.org/10.5194/tc-7-375-2013).
- Independent ice thickness and active-lake inventory: USAP-DC 601470 (Stubblefield et al., 2021, https://doi.org/10.15784/601470; ice thickness after Arthern et al., 2015; active lakes after Siegfried & Fricker, 2018).

- Active subglacial-lake drainage volumes: USAP-DC 601439 (Smith et al., 2012, https://doi.org/10.15784/601439
  ICESat inventory of Smith et al., 2009).
- BedMachine Antarctica v4 (Morlighem, 2025, https://doi.org/10.5067/POJQI54A45HX; see Morlighem et al., 2020) for the independent continental RTN screen.
- ITS_LIVE surface velocities (Gardner et al., 2025, https://doi.org/10.5067/JQ6337239C96; see Gardner et al., 2018) and CryoSat-2 lake-drainage dates.

USAP-DC data were obtained from the U.S. Antarctic Program Data Center (https://www.usap-dc.org); BedMachine and ITS_LIVE from the NASA NSIDC DAAC.

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